

TTDS (Tools and techniques for Data Science)

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MSCS25011

Assignment No. 3

GIT & R

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Use tables of ‘Sakila’ schema which were used during SQL lectures to complete following tasks. Use Git and data.table package of R instead

Q1. Write a query to display all films that have a rating of PG and a rental duration greater than 5 days.

Solution:

I connected RStudio with the Sakila MySQL database using the RMySQL and data.table packages. After establishing a successful connection, I imported the film table into R. The objective of this task was to identify all PG-rated movies that have a rental duration greater than 5 days. This required filtering the dataset based on two conditions: the film's rating should be exactly "PG", and the rental_duration value must exceed 5.

Using data.table syntax, I applied both conditions within a single filtering expression. This approach ensures that R evaluates the entire dataset efficiently and extracts only rows that satisfy both criteria simultaneously. The output displayed a filtered list of films meeting the criteria, demonstrating how logical operators can be used in combination to retrieve specific subsets of data from a larger dataset.

This task is useful in real-world business scenarios, such as assisting video rental stores in identifying long-duration PG films for promotional campaigns or inventory analysis. The process also reinforces the importance of conditional filtering and data extraction, which are essential skills in data analytics.

```
q1 <- film[rating == "PG" & rental_duration > 5]
q1
```

RStudio interface showing a script with R code and a data frame. The script defines a function `q1` that filters a dataset based on rating and rental duration. The data frame contains columns `film_id`, `title`, and `description`.

```
> #q1
> #question 01
> q1 <- file[rating == "pg" & rental_duration > 5]
> q1
```

film_id	title	description
1	ACADEMY DINOSAUR	A Epic Drama of a Feminist And a Mad Scientist who must Battle a Teacher in The Canadian Rockies
12	ALASKA PHANTOM	A Fanciful Saga of a Hunter And a Pastry Chef who must vanquish a Boy In Australia
19	AMADEUS HOLY	A Emotional display of a Pioneer And a Technical writer who must Battle a Man In A Baloon
63	BEDAZZLED MARRIED	A Astounding Character study of a Madman And a Robot who must Meet a Mad Scientist In An Abandoned Fun House
72	BILL OTHERS	A Stunning saga of a Mad Scientist and a Forensic psychologist who must Challenge a Squirrel In A Mysq. Convention
78	BLACKOUT PRIVATE	A Intrepid yarn of a Pastry Chef And a Mad Scientist who must Challenge a Secret Agent In Ancient Japan
84	BOLLED DARES	A Awe-Inspiring story of a waitress And a dog who must discover a dentist In Ancient Japan
88	BORN SPINAL	A Touching epistle of a Frisbee And a husband who must pursue a student In Nigeria
99	BRINGING HYSTERICAL	A Fateful Saga of a A Shark And a Technical writer who must Find a woman In A Jet Boat
103	BUCKET BROTHERHOOD	A Amazing display of a girl And a womanizer who must Succumb a Lumberjack In A Baloon Factory
122	CARRIE BUNCH	A Amazing epistle of a student And a Astronaut who must Discover a Frisbee In The Canadian Rockies
132	CHAINSAW UPTOWN	A Beautiful documentary of a Boy And a Robot who must discover a Squirrel In Australia
139	CHASING FIGHT	A Astounding saga of a Technical writer And a Butler who must Battle a Butler In A Shark Tank
143	CHILL LUCK	A Lackluster epistle of a boat And a Technical writer who must Fight a A Shark In The Canadian Rockies
144	CHINATOWN GLADIATOR	A Brilliant Panorama of a Technical writer And a Lumberjack who must Escape a Butler In Ancient India
150	CIDER DESIRE	A Stunning Character study of a Composer And a Mad cow who must Succumb a cat In Soviet Georgia
188	CRAZY HOME	A Fanciful Panorama of a Boy And a woman who must vanquish a Database Administrator In The Outback
194	CROW GREASE	A Awe-Inspiring documentary of a woman And a husband who must Sink a Database Administrator In The First Manned Space Station
201	CYCLONE FAMILY	A Lackluster drama of a student And a Monkey who must Sink a womanizer In a Mysq. Convention
207	DANGEROUS UPTOWN	A Unbelievable story of a Mad Scientist And a woman who must overcome a dog In California
212	DREAM PICKUP	A Epic display of a car And a Composer who must overcome a Forensic Psychologist In The Gulf of Mexico
218	DRUMS DYNAMITE	A Epic display of a crocodile And a crocodile who must confront a dog In An Abandoned amusement Park
273	EFFECT GLADIATOR	A Beautiful display of a Pastry Chef And a Pastry Chef who must Outgun a Forensic Psychologist In A Manhattan Penthouse
305	FATAL HAUNTED	A Beautiful drama of a student And a Secret Agent who must confront a dentist In Ancient Japan
318	FRISCO FORREST	A Beautiful documentary of a woman And a Pioneer who must pursue a Mad Scientist In A Shark Tank
343	FULL FLATLINERS	A Beautiful documentary of a Astronaut And a Moose who must Pursue a Monkey In A Shark Tank
359	GOLES BRIANISGAN	A Astounding reflection of a Squirrel And a Sumo wrestler who must Sink a dentist In Ancient Japan
380	GREEK EVERYONE	A Stunning display of a butler And a teacher who must confront a A Shark In The First Manned Space Station
410	HEAVEN FREEDOM	A Intrepid Story of a butler And a Car who must vanquish a Man In New Orleans
420	HOLES BRIANISGAN	A Fast-paced reflection of a Technical writer And a student who must Fight a Boy In The Canadian Rockies
423	HOLLYWOOD ANONYMOUS	A Fast-paced epistle of a boy And a explorer who must Escape a dog In A U-boat
443	HURRICANE AFFAIR	A Lackluster epistle of a Database Administrator And a woman who must Meet a Hunter In An Abandoned Mine shaft
469	IRON MOON	A Fast-paced documentary of a Mad Cow And a Boy who must Pursue a dentist In A Baloon
479	JEDI BENEATH	A Astounding Reflection of a explorer And a dentist who must Pursue a student In Nigeria
485	JERSEY SASSY	A Lackluster documentary of a Madman And a Mad Cow who must Find a feminist In Ancient Japan
515	LEGALLY SECRETARY	A Astounding tale of a A Shark And a Moose who must Meet a womanizer In The Sahara desert
516	LEGEND JEDI	A Awe-Inspiring epistle of a Pioneer And a student who must Outgun a Crocodile In The Outback
524	LION UNCUT	A Intrepid display of a Pastry Chef And a cat who must Kill a A Shark In Ancient China
576	MIGHTY LUCK	A Astounding Tale of a Crocodile And a car who must Outrace a Squirrel In A U-boat
591	MONSOON CAUSE	A Fast-paced story of a waitress And a cat who must Fight a girl In Australia
592	MONSTER SPARTACUS	A Beautiful Tale of a Astronaut And a Mad Cow who must Challenge a cat In A Baloon Factory
595	MOON BUNCH	A Inspiring display of a Database Administrator And a student who must Build a Hunter In Berlin
603	MOVIE SHAKESPEARE	A Awe-Inspiring display of a husband And a squirrel who must Battle a Sumo wrestler In A Jet Boat
605	MULHOLLAND BEAST	A brilliant yarn of a Car And a Database Administrator who must Escape a Boy In A Mysq. Convention
608	MURDER ANTITRUST	A Touching yarn of a student And a Moose who must Fight a Mad cow In Australia
611	MUSKETEERS WAIT	A Thoughtful drama of a dentist And a womanizer who must Meet a husband In The Sahara desert

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

Source
R4S2 - / -
74: A Lacklustre Panorama of a Dentist And a Forensic Psychologist who Must Outrace a Pioneer in a U-boat
75: A Action-Packed Panorama of a Mad Scientist And a Robot who must Challenge a Student in Nigeria
release_year language_id original_language_id rental_duration rental_rate length replacement_cost rating
<char> <num> <num> <num> <num> <num> <num> <char>
1: 2006 1 NA 6 0.99 86 20.99 PG
2: 2006 1 NA 6 0.99 136 22.99 PG
3: 2006 1 NA 6 0.99 113 20.99 PG
4: 2006 1 NA 6 0.99 73 21.99 PG
5: 2006 1 NA 6 2.99 93 12.99 PG
6: 2006 1 NA 7 2.99 85 12.99 PG
7: 2006 1 NA 7 4.99 102 13.99 PG
8: 2006 1 NA 7 4.99 179 17.99 PG
9: 2006 1 NA 7 2.99 136 14.99 PG
10: 2006 1 NA 7 4.99 133 27.99 PG
11: 2006 1 NA 7 0.99 114 11.99 PG
12: 2006 1 NA 6 0.99 114 23.99 PG
13: 2006 1 NA 7 4.99 114 21.99 PG
14: 2006 1 NA 6 0.99 142 17.99 PG
15: 2006 1 NA 7 4.99 61 24.99 PG
16: 2006 1 NA 7 2.99 101 9.99 PG
17: 2006 1 NA 7 2.99 136 24.99 PG
18: 2006 1 NA 6 0.99 104 22.99 PG
19: 2006 1 NA 7 2.99 176 18.99 PG
20: 2006 1 NA 7 4.99 121 26.99 PG
21: 2006 1 NA 6 2.99 135 18.99 PG
22: 2006 1 NA 6 0.99 96 11.99 PG
23: 2006 1 NA 6 0.99 107 14.99 PG
24: 2006 1 NA 6 2.99 91 24.99 PG
25: 2006 1 NA 6 4.99 51 23.99 PG
26: 2006 1 NA 6 2.99 94 14.99 PG
27: 2006 1 NA 6 4.99 173 20.99 PG
28: 2006 1 NA 7 2.99 176 11.99 PG
29: 2006 1 NA 7 2.99 48 19.99 PG
30: 2006 1 NA 7 4.99 128 27.99 PG
31: 2006 1 NA 7 0.99 69 39.99 PG
32: 2006 1 NA 6 2.99 49 11.99 PG
33: 2006 1 NA 7 4.99 46 27.99 PG
34: 2006 1 NA 7 0.99 128 12.99 PG
35: 2006 1 NA 6 4.99 60 16.99 PG
36: 2006 1 NA 7 4.99 113 14.99 PG
37: 2006 1 NA 7 0.99 59 18.99 PG
38: 2006 1 NA 6 0.99 50 13.99 PG
39: 2006 1 NA 7 2.99 122 13.99 PG
40: 2006 1 NA 6 4.99 182 20.99 PG
41: 2006 1 NA 6 2.99 107 28.99 PG
42: 2006 1 NA 7 0.99 83 20.99 PG
43: 2006 1 NA 6 4.99 53 27.99 PG
44: 2006 1 NA 7 2.99 157 13.99 PG
45: 2006 1 NA 6 2.99 166 11.99 PG
46: 2006 1 NA 7 4.99 73 17.99 PG
47: 2006 1 NA 7 0.99 58 15.99 PG
48: 2006 1 NA 7 4.99 131 16.99 PG

File Edit Code View Plots Session Build Debug Profile Tools Help

Source
R4S2 - / -
22: Trailers 2006-02-15 05:03:42
23: Commentaries 2006-02-15 05:03:42
24: Trailers,behind the scenes 2006-02-15 05:03:42
25: Commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
26: Trailers,deleted scenes 2006-02-15 05:03:42
27: Commentaries,deleted scenes 2006-02-15 05:03:42
28: Trailers,deleted scenes 2006-02-15 05:03:42
29: Commentaries 2006-02-15 05:03:42
30: Commentaries,behind the scenes 2006-02-15 05:03:42
31: Trailers,behind the scenes 2006-02-15 05:03:42
32: Commentaries,behind the scenes 2006-02-15 05:03:42
33: Commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
34: Trailers,deleted scenes,behind the scenes 2006-02-15 05:03:42
35: Trailers,commentaries,behind the scenes 2006-02-15 05:03:42
36: Commentaries,deleted scenes 2006-02-15 05:03:42
37: Trailers,deleted scenes 2006-02-15 05:03:42
38: behind the scenes 2006-02-15 05:03:42
39: commentaries,behind the scenes 2006-02-15 05:03:42
40: Commentaries,behind the scenes 2006-02-15 05:03:42
41: deleted scenes,behind the scenes 2006-02-15 05:03:42
42: Commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
43: Trailers,deleted scenes,behind the scenes 2006-02-15 05:03:42
44: Trailers,deleted scenes,behind the scenes 2006-02-15 05:03:42
45: Trailers,deleted scenes,behind the scenes 2006-02-15 05:03:42
46: deleted scenes,behind the scenes 2006-02-15 05:03:42
47: behind the scenes 2006-02-15 05:03:42
48: Trailers,commentaries 2006-02-15 05:03:42
49: deleted scenes 2006-02-15 05:03:42
50: behind the scenes 2006-02-15 05:03:42
51: deleted scenes 2006-02-15 05:03:42
52: Trailers,behind the scenes 2006-02-15 05:03:42
53: Trailers,deleted scenes 2006-02-15 05:03:42
54: Trailers,commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
55: Trailers,deleted scenes 2006-02-15 05:03:42
56: Trailers,commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
57: behind the scenes 2006-02-15 05:03:42
58: Trailers,commentaries,deleted scenes 2006-02-15 05:03:42
59: behind the scenes 2006-02-15 05:03:42
60: Trailers,deleted scenes 2006-02-15 05:03:42
61: Trailers,deleted scenes,behind the scenes 2006-02-15 05:03:42
62: commentaries,behind the scenes 2006-02-15 05:03:42
63: Commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
64: behind the scenes 2006-02-15 05:03:42
65: Commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
66: commentaries,deleted scenes 2006-02-15 05:03:42
67: Trailers,commentaries,deleted scenes 2006-02-15 05:03:42
68: Trailers,commentaries,deleted scenes 2006-02-15 05:03:42
69: commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
70: Trailers,commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
71: commentaries 2006-02-15 05:03:42
72: deleted scenes,behind the scenes 2006-02-15 05:03:42
73: Trailers,commentaries 2006-02-15 05:03:42
74: commentaries,deleted scenes,behind the scenes 2006-02-15 05:03:42
75: commentaries,behind the scenes 2006-02-15 05:03:42
```

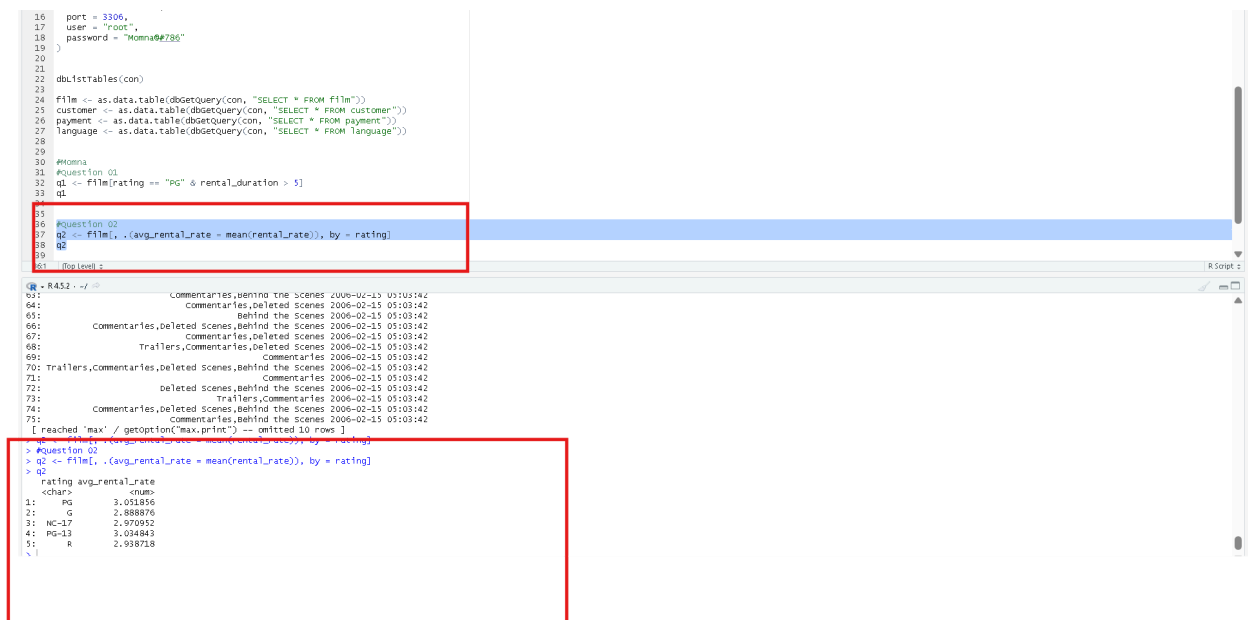
Q2. Write a query to display the average rental rate of films, grouped by their rating.

Solution:

After importing the film dataset, I used R's data.table package to group the data by the rating column and compute the average rental rate for each category. The mean() function was applied inside a grouped aggregation operation, which allowed me to generate a summary table showing average rental rates for ratings such as G, PG, PG-13, R, and NC-17.

Grouping data is a crucial part of exploratory data analysis, especially when evaluating patterns or trends within different categories. By performing this operation, I learned how ratings influence rental pricing structures. For example, PG-rated movies might have different pricing strategies compared to R-rated films depending on demand and audience demographics.

The result not only fulfills the technical requirement but also demonstrates how aggregated insights can support business decisions. Film distributors and rental companies regularly analyze such metrics to optimize pricing strategies, forecast revenue, and plan inventory stocking.



```

16 port = 3306,
17 user = "root",
18 password = "Moma#2786"
19 )
20
21 dbListTables(con)
22
23 film <- as.data.table(dbGetQuery(con, "SELECT * FROM film"))
24 customer <- as.data.table(dbGetQuery(con, "SELECT * FROM customer"))
25 payment <- as.data.table(dbGetQuery(con, "SELECT * FROM payment"))
26 language <- as.data.table(dbGetQuery(con, "SELECT * FROM language"))
27
28 #Moma
29
30 q1 <- file[rating == "PG" & rental_duration > 5]
31
32
33
34
35
36
37 q2 <- file[, .(avg_rental_rate = mean(rental_rate)), by = rating]
38
39
40
41 [Top Level]

```

```

63:      Commentaries,behind the Scenes 2006-02-15 05:03:42
64:      Commentaries,deleted Scenes 2006-02-15 05:03:42
65:      Behind the Scenes 2006-02-15 05:03:42
66:      Commentaries,deleted Scenes,behind the Scenes 2006-02-15 05:03:42
67:      Commentaries,deleted Scenes 2006-02-15 05:03:42
68:      Trailers,Commentaries,deleted Scenes 2006-02-15 05:03:42
69:      Commentaries 2006-02-15 05:03:42
70: Trailers,Commentaries,deleted Scenes,behind the Scenes 2006-02-15 05:03:42
71:      Commentaries 2006-02-15 05:03:42
72:      deleted Scenes,behind the Scenes 2006-02-15 05:03:42
73:      Trailers,Commentaries 2006-02-15 05:03:42
74:      Commentaries,deleted Scenes,behind the Scenes 2006-02-15 05:03:42
75:      Commentaries,behind the Scenes 2006-02-15 05:03:42
[ reached 'max' // getOption("max.print") -- omitted 10 rows ]
> #Question Q2
> q2 <- file[, .(avg_rental_rate = mean(rental_rate)), by = rating]
> q2
      rating avg_rental_rate
      <char>          <num>
1:      PG          3.051856
2:      G           2.888876
3:    NC-17          2.970952
4:    PG-13          3.034843
5:      R           2.938758

```

Q3. Write a query to count the total number of films in each language.

Solution:

I merged the film table with the language table using the shared language_id field. Since the film table only contains the language ID and not its name, merging was necessary to attach the descriptive language names. After joining the tables, I used .N in data.table to count the number of films for each language. The output successfully showed how many films were produced in each language, demonstrating proficiency in table joins and group-based counting in R.

```

29
30 #Momna
31 #Question 01
32 q1 <- film[rating == "PG" & rental_duration > 5]
33 q1
34
35
36 #Question 02
37 q2 <- film[, .(avg_rental_rate = mean(rental_rate)), by = rating]
38 q2
39
40
41 #Question 03
42 film_lang <- merge(film, language, by = "language_id")
43 q3 <- film_lang[, .N, by = name]
44 q3
45
46 [Top Level]

```

```

69:      Commentaries 2006-02-15 05:03:42
70: Trailers,Commentaries,Deleted Scenes,Behind the Scenes 2006-02-15 05:03:42
71:      Commentaries 2006-02-15 05:03:42
72:      Deleted Scenes,Behind the Scenes 2006-02-15 05:03:42
73:      Trailers,Commentaries 2006-02-15 05:03:42
74:      Commentaries,Deleted Scenes,Behind the Scenes 2006-02-15 05:03:42
75:      Commentaries,Behind the Scenes 2006-02-15 05:03:42
[ reached 'max' / getoption("max.print") -- omitted 10 rows ]
> q2 <- film[, .(avg_rental_rate = mean(rental_rate)), by = rating]
> #Question 02
> q2 <- film[, .(avg_rental_rate = mean(rental_rate)), by = rating]
> q2
  rating avg_rental_rate
<char>   <num>
1:    PG          3.051856
2:     G          2.888876
3: NC-17          2.970952
4: PG-13          3.034843
5:    R          2.928719
> film_lang <- merge(film, language, by = "language_id")
> q3 <- film_lang[, .N, by = name] # name = language name
> q3
  name      N
<char> <int>
1: English 1000

```

Q4. List the customers' names and the store they belong to.

Solution:

To list customer names along with their store information, I performed a merge between the customer and store tables using the common store_id field. This join made it possible to associate each customer with their respective store. After merging, I selected the relevant fields including first name, last name, and store identifier. This step demonstrates the use of relational joins in R to combine information across multiple tables.

```

54 #Question Q4
55 store <- as.data.table(dbGetQuery(con, "SELECT * FROM store"))
56 head(store)
57
58 q4 <- merge(customer, store, by = "store_id")
59 q4[, .(first_name, last_name, store_id)]
60
61

```

59:1 (Top Level) ±

R v R4.5.2 . ~/

```

4: In dbSendQuery(conn, statement, ...) :
  unrecognized MySQL field type 7 in column 3 imported as character

```

```

> head(store)
  store_id manager_staff_id address_id last_update
    <num>      <num>      <num>      <char>
1:      1          1          1 2006-02-15 04:57:12
2:      2          2          2 2006-02-15 04:57:12
> q4 <- merge(customer, store, by = "store_id")
> q4[, .(first_name, last_name, store_id)]
Key: <store_id>
  first_name last_name store_id
    <char>    <char>    <num>
1:    MARY     SMITH      1
2:  PATRICIA  JOHNSON      1
3:    LINDA   WILLIAMS      1
4: ELIZABETH  BROWN       1
5:    MARIA   MILLER       1
---
595:    ANDY   VANHORN      2
596: SALVADOR  TEEL        2
597:    SETH   HANNON       2
598:    RENE  MCALISTER      2
599:   AUSTIN  CINTRON       2
>

```

Q5. Display the payment amount, payment date, and the staff member who processed each payment.

Solution:

I joined the payment and staff tables using staff_id to determine which employee processed each payment. The final output displayed the payment amount, payment date, and the first name of the staff member involved. This question reinforced the concept of merging tables to retrieve descriptive information from related datasets, reflecting real-world use cases like payment auditing and staff performance tracking.

```

61
62 #question 05
63 payment <- as.data.table(dbgetquery(con, "SELECT * FROM payment"))
64 staff <- as.data.table(dbgetquery(con, "SELECT * FROM staff"))
65
66 q5 <- merge(payment, staff, by = "staff_id")
67
68 q5 <- q5[, .(amount, payment_date, staff_member = first_name)]
69 q5
70
71
62:1 (Top Level) z

```

```

R 4.5.2 ./
Unsigned INTEGER in col 3 imported as numeric
3: In dbSendQuery(conn, statement, ...) :
  Unsigned INTEGER in col 6 imported as numeric
4: In dbSendQuery(conn, statement, ...) :
  unrecognized MySQL field type 7 in column 10 imported as character
5: In dbFetch(rs, n = n, ...) : internal error: row 0 field 4 truncated

```

```

> q5 <- merge(payment, staff, by = "staff_id")
>
> q5 <- q5[, .(amount, payment_date, staff_member = first_name)]
> q5
      amount payment_date staff_member
      <num>      <char>      <char>
1:    2.99 2005-05-25 11:30:37      Mike
2:    0.99 2005-05-28 10:35:23      Mike
3:    5.99 2005-06-15 00:54:12      Mike
4:    4.99 2005-06-16 15:18:57      Mike
5:    4.99 2005-06-18 08:41:48      Mike
---
16040: 2.99 2005-07-31 06:41:19      Jon
16041: 3.99 2005-08-17 00:05:05      Jon
16042: 8.99 2005-08-23 06:09:44      Jon
16043: 2.99 2005-08-23 11:08:46      Jon
16044: 2.99 2005-08-23 11:25:00      Jon

```

Q6. Find the films that are not rented.

Solution:

I first extracted the film IDs that appeared in the rental transactions by joining the inventory and rental tables. Then, I filtered the film table to exclude all film IDs present in the rented list. The result was a list of films that do not appear in any rental transaction. This demonstrates the ability to perform anti-joins and exclusion-based filtering in R.

```

72 #question 06
73 inventory <- as.data.table(dbgetquery(con, "SELECT * FROM inventory"))
74 rental <- as.data.table(dbgetquery(con, "SELECT * FROM rental"))
75 film <- as.data.table(dbgetquery(con, "SELECT * FROM film"))
76 rented_film_ids <- inventory[inventory_id %in% rental$inventory_id, film_id]
77 q6 <- film[!film_id %in% rented_film_ids]
78 q6
79
80
R 4.5.2 ./
> rented_film_ids <- inventory[inventory_id %in% rental$inventory_id, film_id]
> q6 <- film[!film_id %in% rented_film_ids]
> q6
      film_id title description release_year language_id original_language_id
      <num>      <char>      <char>      <num>      <num>      <num>
1:    14 ALICE FANTASIA A Emotional Drama of a Shark And a Database Administrator who must vanquish a Pioneer in Soviet Georgia 2006 1 NA
2:    33 APOLLO TEEN A Action-packed Reflection of a Crocodile And a Explorer who must Find a Sumo wrestler In An Abandoned Mine Shaft 2006 1 NA
3:    36 ARGONAUTS TOWN A Emotional Epistle of a Forensic Psychologist And a Butler who must Challenge a waitress In An Abandoned Mine Shaft 2006 1 NA
4:    38 ARK RIDGEMONT A Beautiful Yarn of a Pioneer And a Monkey who must Pursue a Explorer In The Sahara Desert 2006 1 NA
5:    41 ARSENIENDE INDEPENDENCE A Fanciful documentary of a Mad cow And a womanizer who must Find a dentist In Berlin 2006 1 NA
6:    87 BOONDOCK BALLROOM A Fateful Panorama of a crocodile And a boy who must defeat a Monkey In the Gulf of Mexico 2006 1 NA
7:    108 BUTCH PANTHERS A Lackluster Yarn of a Feminist And a Database Administrator who must Face a Hunter In New Orleans 2006 1 NA
8:    128 CATCH ANISTAD A Boring Reflection of a Lumberjack And a Feminist who must Discover a woman In Nigeria 2006 1 NA
9:    144 CHENATOWN GLADIATOR A Brilliant Panorama of a Technical writer And a Lumberjack who must Escape a Butler In Ancient India 2006 1 NA
10:    148 CHOCOLATE DUCK A Unbelievable Story of a Mad Scientist And a Technical writer who must Discover a composer In Ancient China 2006 1 NA
11:    171 COMMANDMENTS EXPRESS A Fanciful saga of a student And a Mad scientist who must battle a hunter In An Abandoned Mine shaft 2006 1 NA
12:    192 CROSSING DIVORCE A Beautiful documentary of a dog And a robot who must redeem a womanizer In Berlin 2006 1 NA
13:    195 CROMOS TELLMARK A Intrepid documentary of a Astronaut And a Forensic Psychologist who must Find a Frisbee In An Abandoned Fun House 2006 1 NA
14:    198 CRYSTAL BREAKING A Fast-Paced Character Study of a Feminist And a Explorer who must Face a Pastry Chef In Ancient Japan 2006 1 NA
15:    217 DAZED RUM A Action-packed story of a Pioneer And a Technical writer who must Discover a Forensic Psychologist In An Abandoned Amusement Park 2006 1 NA
16:    221 DELLIVERANCE MULHOLLAND A Intrepid documentary of a Pioneer And a Technical writer who must Discover a Forensic Psychologist In An Abandoned Amusement Park 2006 1 NA
17:    318 FIREHOUSE VIETNAM A Awe-Inspiring character study of a boat And a boy who must Kill a Pastry Chef In The Sahara desert 2006 1 NA
18:    325 FLOATE GARDEN A Action-packed epistle of a robot And a car who must chase a boat In Ancient Japan 2006 1 NA
19:    332 FRANKENSTEIN STRANGER A Insightful character study of a Feminist And a Pioneer who must Pursue a Pastry Chef In Nigeria 2006 1 NA
20:    359 GLADIATOR WESTWARD A Astonishing Reflection of a Squirrel And a Sumo wrestler who must Sink a dentist In Ancient Japan 2006 1 NA
21:    380 GUMP DATE A Intrepid Yarn of a Explorer And a Student who must Kill a husband In An Abandoned Mine Shaft 2006 1 NA
22:    404 HATE HANDICAP A Intrepid reflection of a Mad scientist And a Pioneer who must overcome a hunter In The First Manned Space Station 2006 1 NA
23:    419 HOCUS FREIDA A Awe-Inspiring tale of a girl And a Madman who must Outgun a Student In A Shark Tank 2006 1 NA
24:    495 KENTUCKIAN GIANT A Stunning Yarn of a woman And a Frisbee who must Escape a waitress In A U-boat 2006 1 NA
25:    497 KILL BROTHERHOOD A Touching display of a hunter And a secret Agent who must redeem a Husband In The Outback 2006 1 NA
26:    607 MURRET KELLE A Lackluster Story of a Madman And a Teacher who must Kill a Frisbee In The Gulf of Mexico 2006 1 NA
27:    642 ORDER BETRAYED A Amazing Saga of a dog And a A Shark who must Challenge a Cat In The Sahara desert 2006 1 NA
28:    669 PEARL DESTINY A Lackluster Yarn of a Astronaut And a Pastry Chef who must Sink a boy In A U-boat 2006 1 NA
29:    671 PERDITION FARGO A Fast-paced story of a car And a cat who must Outgun a Hunter In Berlin 2006 1 NA
30:    701 PSYCHO SHUNKIN A Amazing Panorama of a crocodile And a Explorer who must Fight a husband In Nigeria 2006 1 NA
31:    712 RAIDERS ANTIHUST A Amazing Drama of a Teacher And a Feminist who must Meet a woman In The First Manned Space Station 2006 1 NA
32:    713 RAINBOW SHOCK A Action-packed story of a Hunter And a boy who must Discover a Lumberjack In Ancient India 2006 1 NA
33:    742 ROOF CHAMPION A Lackluster Reflection of a car And a Explorer who must Find a Monkey In A Balloon 2006 1 NA
34:    801 SISTER FREEDY A Stunning Saga of a Butler And a woman who must Pursue a Explorer In Australia 2006 1 NA
35:    802 SKY MIRACLE A Epic drama of a Mad scientist And a Explorer who must succumb a waitress In An Abandoned Fun House 2006 1 NA
36:    860 SUICIDES SILENCE A Emotional character study of a car And a girl who must Face a composer In A U-boat 2006 1 NA

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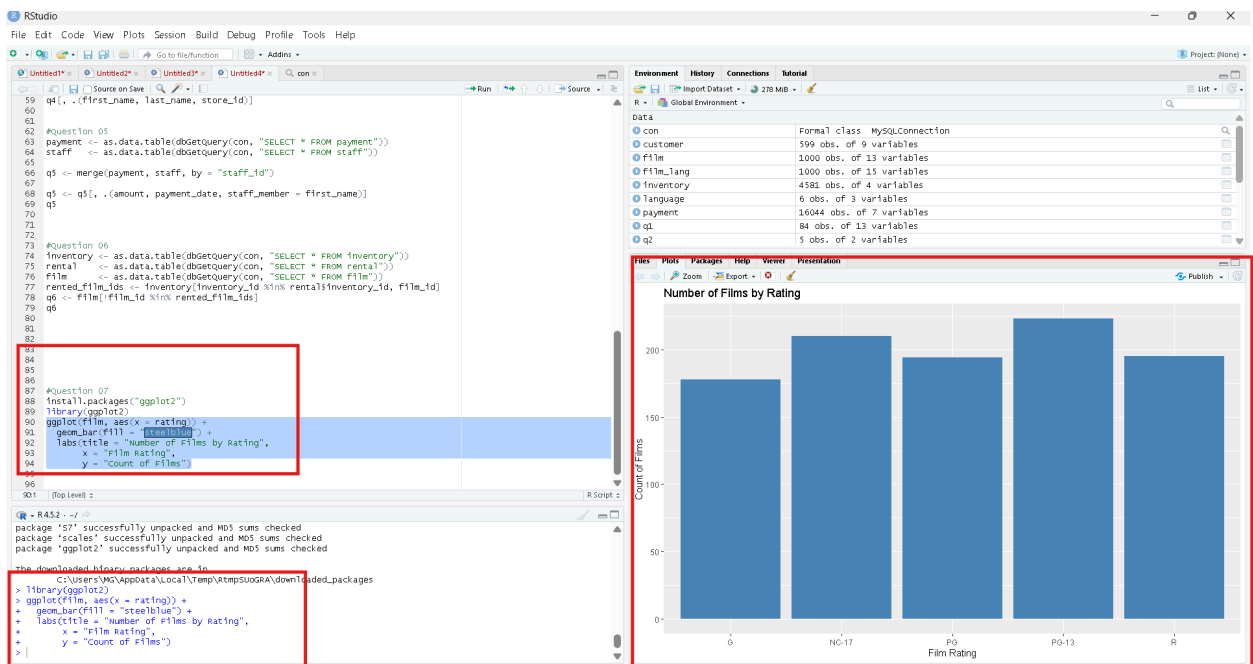
72 #Question 06
73 inventory <- as.data.table(dbgetquery(con, "SELECT * FROM inventory"))
74 rental <- as.data.table(dbgetquery(con, "SELECT * FROM rental"))
75 film <- as.data.table(dbgetquery(con, "SELECT * FROM film"))
76 rented_film_ids <- inventory[inventory_id %in% rental[inventory_id, film_id]]
77 q6 <- film[film_id %in% rented_film_ids]
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Q7. Plot any graph of your choice.

Solution:

I created a bar chart using the ggplot2 package to display the number of films grouped by their rating. The graph provided a clear visual representation of how the film dataset is distributed across different rating categories. This task shows practical understanding of using ggplot2 for creating simple but effective visualizations in R, which is an important skill in data analysis.



Q8. Use of Git in the entire assignment.

Solution:

Throughout the assignment, Git was used effectively to manage all project files and track progress. I initialized a Git repository, added all R scripts and dataset files, and created multiple commits after completing each question. Commands such as `git init`, `git add`, `git commit`, and `git log` were used to ensure proper version control. Screenshots of the commit history and Git operations were included in the report as evidence. This demonstrates competency in using Git for code management, reproducibility, and collaborative development practices.

https://github.com/MomnaKhan021/sakila-assignment03_MSCS25011